Kamino DVI Solver Benchmark

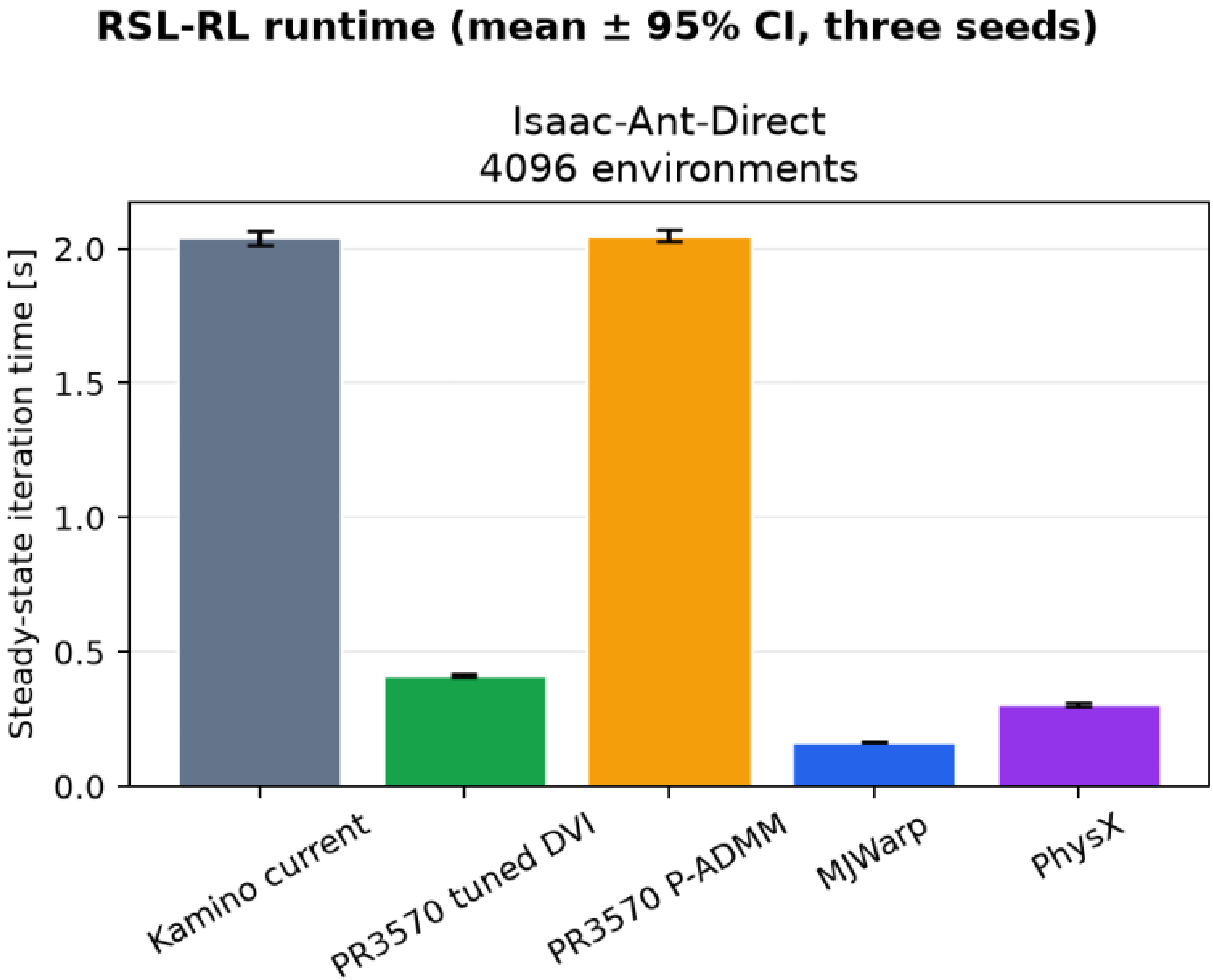
RSL-RL · 300 iterations · three seeds · mean ± 95% Student-t CI

- Isaac-Ant-Direct: tuned DVI is 4.9× faster than current Kamino and 5.0× faster than PR3570 P-ADMM.
- Isaac-Ant-Direct: tuned DVI remains 2.5× slower than MJWarp and 1.4× slower than PhysX.

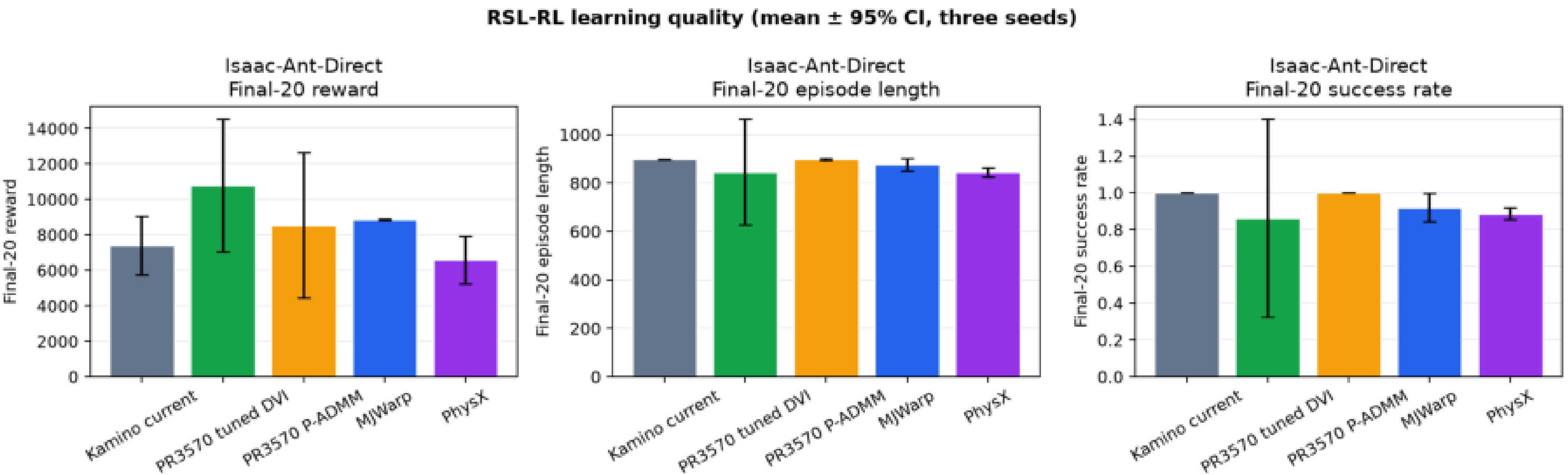
Task	Variant	Envs	Iteration [s]	Total FPS	Reward
Isaac-Ant-Direct	Kamino current	4096	2.040 ± 0.028	64263 ± 890	7402.741 ± 1650.401
Isaac-Ant-Direct	PR3570 tuned DVI	4096	0.413 ± 0.005	317194 ± 3709	10785.473 ± 3745.488
Isaac-Ant-Direct	PR3570 P-ADMM	4096	2.048 ± 0.023	63990 ± 722	8547.088 ± 4094.777
Isaac-Ant-Direct	MJWarp	4096	0.167 ± 0.001	785924 ± 4857	8872.347 ± 33.530
Isaac-Ant-Direct	PhysX	4096	0.304 ± 0.007	433084 ± 9769	6588.875 ± 1341.016

- Schema v1.1 success series differs from TensorBoard in 15 of 15 runs; report uses TensorBoard success. This is the known generator bug addressed separately in PR 6624.
- Isaac-Ant-Direct tuned DVI success is seed-sensitive: the three-seed 95% CI half-width is 0.539.

Runtime


RSL-RL runtime (mean ± 95% CI, three seeds)
Isaac-Ant-Direct
4096 environments
Steady-state iteration time [s]
0.0
0.5
1.0
1.5
2.0
Kamino current
PR3570 tuned DVI
PR3570 P-ADMM
MJWarp
PhysX

Learning


RSL-RL learning quality (mean ± 95% CI, three seeds)
Isaac-Ant-Direct
Final-20 reward
Final-20 reward
0
2000
4000
6000
8000
10000
12000
14000
Isaac-Ant-Direct
Final-20 episode length
Final-20 episode length
0
200
400
600
800
1000
Isaac-Ant-Direct
Final-20 success rate
Final-20 success rate
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
Kamino current
PR3570 tuned DVI
PR3570 P-ADMM
MJWarp
PhysX